

Topology and Manifolds

MH3600

Chapter 3: Continuity and Metric Spaces

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Outline

- 1 Continuous functions
- 2 The metric topology
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Continuous functions

Continuity is a **fundamental concept in topology**. For example, when you hear that “a coffee cup and a doughnut are topologically equivalent”, this is really a claim about the existence of a certain continuous function. Recall that the standard definition of **continuity** of a **real valued function of a real variable** at a point $x_0 \in \mathbb{R}$ in the domain $\mathcal{D}(f) \subset \mathbb{R}$ of the function $f : \mathcal{D}(f) \subset \mathbb{R} \longrightarrow \mathbb{R}$, is as follows:

Definition

Suppose $f : \mathcal{D}(f) \subset \mathbb{R} \longrightarrow \mathbb{R}$ is a function and $x_0 \in \mathcal{D}(f)$. Then f is **continuous at point x_0** if for all $\varepsilon > 0$ there exists $\delta(\varepsilon) > 0$ such that for all $|x - x_0| < \delta(\varepsilon)$ and $x \in \mathcal{D}(f)$ we have $|f(x) - f(x_0)| < \varepsilon$.

We then say that f is **continuous at a set $A \subset \mathbb{R}$** if f is continuous at each point of A .

The following is a consequence of the previous definition:

Theorem

Let $f : \mathcal{D}(f) \subset \mathbb{R} \rightarrow \mathbb{R}$. Then f is continuous on $\mathcal{D}(f)$ **if and only if** for all open sets $V \subset \mathbb{R}$, we have $f^{-1}(V)$ is open relative to $\mathcal{D}(f)$.

Inspired by this result, we have the following as our (standard) definition of continuous functions. We take functions defined on all of topological space X , so there is no need for the “open relative to” part of the above definition.

Definition

Let X and Y be topological spaces. A function $f : X \rightarrow Y$ is **continuous** if for each open subset V of Y , the set $f^{-1}(V)$ is open in X . The symbol $\mathcal{C}(X, Y)$ denotes the set of continuous functions between the topological spaces X and Y .

In Calculus 1, continuity is defined based on limits (which are defined using ϵ 's and δ 's), however we have not yet defined the limit of a function (though we have defined limit points of sets). Our definition is based entirely on open sets and we should in fact state that f is continuous **relative to the topologies on X and Y** .

Lemma

Let $f : X \rightarrow Y$, let $\mathcal{B}$ be a basis for the topology on Y , and let $\mathcal{S}$ be a subbasis for the topology on Y .

- (1)** *f is continuous if $f^{-1}(B)$ is open in X for each $B \in \mathcal{B}$.*
- (2)** *f is continuous if $f^{-1}(S)$ is open in X for each $S \in \mathcal{S}$.*

Proof of the lemma

(1) Let $V \subset Y$ be open. Then (by definition of basis) there are $B_\alpha \in \mathcal{B}$ for $\alpha \in J$ such that $V = \bigcup_{\alpha \in J} B_\alpha$. Then $f^{-1}(V) = f^{-1}(\bigcup_{\alpha \in J} B_\alpha) = \bigcup_{\alpha \in J} f^{-1}(B_\alpha)$ is an open set in X by hypothesis. So each $f^{-1}(B_\alpha)$ is open in X and $f^{-1}(V)$ is open in X . Hence f is continuous.

(2) Let $V \subset Y$ be open. Then (by the definition of subbasis) there are S_α^i for $\alpha \in J, i \in \mathbb{N}$ such that $V = \bigcup_{\alpha \in J} (S_\alpha^1 \cap S_\alpha^2 \cap \cdots \cap S_\alpha^{n_\alpha})$. Then

$$\begin{aligned} f^{-1}(V) &= f^{-1}(\bigcup_{\alpha \in J} (S_\alpha^1 \cap S_\alpha^2 \cap \cdots \cap S_\alpha^{n_\alpha})) = \bigcup_{\alpha \in J} f^{-1}(S_\alpha^1 \cap S_\alpha^2 \cap \cdots \cap S_\alpha^{n_\alpha}) \\ &= \bigcup_{\alpha \in J} (f^{-1}(S_\alpha^1) \cap f^{-1}(S_\alpha^2) \cap \cdots \cap f^{-1}(S_\alpha^{n_\alpha})) \end{aligned}$$

is open in X since each $f^{-1}(S_\alpha^i)$ is open in X by hypothesis and so $f^{-1}(S_\alpha^1) \cap f^{-1}(S_\alpha^2) \cap \cdots \cap f^{-1}(S_\alpha^{n_\alpha})$ is open for each $\alpha \in J$, and hence the union is open. So $f^{-1}(V)$ is open and f is continuous. ■

The new continuity definition and the old one

Example

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ (with $\mathbb{R}$ having the standard topology) be **continuous** according to our new definition. We now see that this definition implies the one based on ε - δ that you learnt in calculus.

Let $x_0 \in \mathbb{R}$ and $\varepsilon > 0$. Then the interval $V = (f(x_0) - \varepsilon, f(x_0) + \varepsilon)$ is open in the range space $\mathbb{R}$. Therefore, $f^{-1}(V)$ is open in the domain space $\mathbb{R}$. Now $x_0 \in f^{-1}(V)$ so there is some basis element (a, b) containing x_0 with $(a, b) \subset f^{-1}(V)$ (recall that the standard topology on $\mathbb{R}$ is defined as the topology with basis $\{(a, b) \mid a, b \in \mathbb{R}, a < b\}$). Choose $\delta = \min \{x_0 - a, b - x_0\}$. Then for $x \in (x_0 - \delta, x_0 + \delta)$ (that is, for $|x - x_0| < \delta$) we have $f(x) \in V$ (that is, $|f(x) - f(x_0)| < \varepsilon$).

So our definition of continuity implies the ε - δ definition of continuity. The converse implication is carried out in the tutorials.

Theorem

Let X and Y be topological spaces. let $f : X \rightarrow Y$. Then the following are equivalent:

- (1) f is continuous.
- (2) For every subset A of X , one has $f(\bar{A}) \subset \overline{f(A)}$.
- (3) For every closed subset B of Y , the set $f^{-1}(B)$ is closed in X .
- (4) For each $x \in X$ and each neighborhood V of $f(x)$, there is a neighborhood U of x such that $f(U) \subset V$.

Homeomorphisms

Definition

Let X and Y be topological spaces. Let $f : X \rightarrow Y$ be a bijection (one to one and onto). If both f and $f^{-1} : Y \rightarrow X$ are continuous, then f is a **homeomorphism**.

If $f : X \rightarrow Y$ is a homeomorphism then U is open in X if and only if $f(U)$ is open in Y . Any property in X that is expressed entirely in terms of the topology on X yields through the homeomorphism the corresponding property in Y . Such a property is called a **topological property** of X . Examples of such properties are **open/closed**, **limit points of a set**, **limits of a sequence**, **a basis or subbasis for the topology**, and as we will see later on, **connectedness and compactness**.

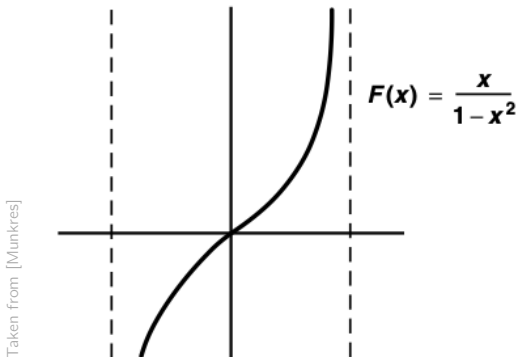
Topological embeddings

Definition

Let $f : X \rightarrow Y$ be an injective (one to one) continuous map. Let $Z = f(X)$ (so that f is onto Z) be considered a subspace of Y . Let $f' : X \rightarrow Z$ be the restriction of f to Z (so f' is a bijection). If f' is a homeomorphism of X with Z , then $f : X \rightarrow Y$ is a **topological embedding** (or simply **embedding**) of X in Y .

Example

Consider $F : (-1, 1) \rightarrow \mathbb{R}$ defined by $F(x) = x / (1 - x^2)$. Then F is continuous and one to one – since $F'(x) = (1 + x^2) / (1 - x^2)^2 \geq 0$ – and is continuous on $\mathbb{R}$. So F is a **homeomorphism** (where $\mathbb{R}$ has the standard topology and $(-1, 1)$ has the subspace topology).



Example

We give an example of a continuous bijective function which has an inverse which is **not continuous**. Consider $[0, 1)$ with the subspace topology (as a subspace of $\mathbb{R}$ with the standard topology) and $S^1 = \{(x, y) \mid x^2 + y^2 = 1\}$ with the subspace topology (as a subspace of $\mathbb{R}^2$ with the standard topology).

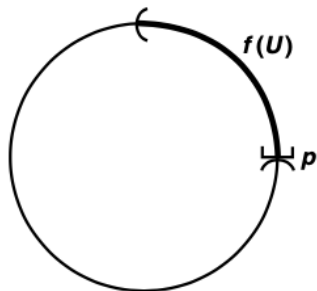
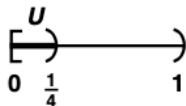
Define $f : [0, 1) \rightarrow S^1$ as $f(t) = (\cos 2\pi t, \sin 2\pi t)$.

Then f is a continuous bijection but f^{-1} is not continuous since

$$f([0, 1/2)) = \{(\cos 2\pi t, \sin 2\pi t) \mid t \in [0, 1/2)\}$$

and so the inverse image (under mapping f ; that is, $(f^{-1})^{-1}([0, 1/2)) = f([0, 1/2))$) of the open set $[0, 1/2)$ is not open.

Taken from [Munkres]



Rules for Constructing Continuous Functions

Theorem

Let X, Y , and Z be topological spaces.

- (a) *(Constant Function)* If $f : X \rightarrow Y$ maps all of X into a single point $y_0 \in Y$, then f is continuous.
- (b) *(Inclusion)* if A is a subspace of X , the inclusion function $j : A \rightarrow X$ is continuous.
- (c) *(Composites)* If $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ are continuous, then the map $g \circ f : X \rightarrow Z$ is continuous.
- (d) *(Restricting the Domain)* If $f : X \rightarrow Y$ is continuous and if A is a subspace of X , then the restricted function $f|_A : A \rightarrow Y$ is continuous.

Note. The following result, which is fairly easy to prove, will be extremely useful in the future.

Theorem (Continued)

- (e) **(Restricting or Expanding the Range)** let $f : X \rightarrow Y$ be continuous. If Z is a subspace of Y containing the image set $f(X)$, then the function $g : X \rightarrow Z$ obtained by restricting the range of f is continuous.
- If Z is a space having Y as a subspace, then the functions $h : X \rightarrow Z$ obtained by expanding the range of f is continuous.
- (f) **(Local Formulation of Continuity)** The map $f : X \rightarrow Y$ is continuous if X can be written as the union of open sets U_α such that $f|_{U_\alpha}$ is continuous for each α .
- (g) **(The Pasting Lemma)** Let $X = A \cup B$, where A and B are closed in X . Let $f : A \rightarrow Y$ and $g : B \rightarrow Y$ be continuous. If $f(x) = g(x)$ for every $x \in A \cap B$, then f and g combine to give a continuous function $h : X \rightarrow Y$, defined by setting $h(x) = f(x)$ if $x \in A$, and $h(x) = g(x)$ if $x \in B$. This result also holds if A and B are open sets in X .

Maps into products

Theorem

Let $f : A \rightarrow X \times Y$ be given by the equation $f(a) = (f_1(a), f_2(a))$ where $f_1 : A \rightarrow X$ and $f_2 : A \rightarrow Y$. Then f is continuous if and only if the functions f_1 and f_2 are continuous.

Definition

The functions $f_1 : A \rightarrow X$ and $f_2 : A \rightarrow Y$, where $f_1 = \pi_1 \circ f$ and $f_2 = \pi_2 \circ f$ for $f : A \rightarrow X \times Y$ above, are called **coordinate functions** of f .

Proof of the theorem

Let $\pi_1 : X \times Y \rightarrow X$ and $\pi_2 : X \times Y \rightarrow Y$. Then for U open in X and V open in Y , we have $\pi_1^{-1}(U) = U \times Y$ and $\pi_2^{-1}(V) = X \times V$ open in $X \times Y$ (by the definition of product topology; these are basis elements for the product topology on $X \times Y$). So π_1 and π_2 are continuous. Note that for each $a \in A$, $\pi_1(f(a)) = \pi_1((f_1(a), f_2(a))) = f_1(a)$ and $\pi_2(f(a)) = \pi_2((f_1(a), f_2(a))) = f_2(a)$. So $f_1 = \pi_1 \circ f$ and $f_2 = \pi_2 \circ f$.

Suppose f is continuous. Then, by Theorem part (c) slide 15, f_1 and f_2 are continuous.

Conversely, suppose f_1 and f_2 are continuous. Let $U \times V$ be a basis element for the product topology of $X \times Y$ (so U is open in X and V is open in Y). Now $a \in f^{-1}(U \times V)$ if and only if $f(a) \in U \times V$, or if and only if $f_1(a) \in U$ and $f_2(a) \in V$, or if and only if $a \in f_1^{-1}(U) \cap f_2^{-1}(V)$. That is, $f^{-1}(U \times V) = f_1^{-1}(U) \cap f_2^{-1}(V)$. Since f_1 and f_2 are continuous then $f_1^{-1}(U)$ and $f_2^{-1}(V)$ are open in X and so $f^{-1}(U \times V)$ is open in X . By Lemma (1) on slide 6, f is continuous.

Remark (Continuous maps on Cartesian products)

There is no useful criterion for the continuity of a map $f : A \times B \rightarrow X$ whose domain is a product space. For example, we might expect that if $f : \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ is continuous if $f(x, y_0)$ and $f(x_0, y)$ are continuous for each $y_0 \in \mathbb{R}$ and $x_0 \in \mathbb{R}$ (that is, if f is continuous "in each variable separately") then $f(x, y)$ is continuous.

This is not the case!

Example:

$$f(x, y) = \begin{cases} xy / (x^2 + y^2) & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } (x, y) = (0, 0) \end{cases}$$

This claim is justified in the tutorial.

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The metric topology

The topological concepts you encounter in the calculus and analysis classes are based on the **metric** on $\mathbb{R}$ which gives the **distance** between x and y in $\mathbb{R}$ as $|x - y|$. More generally, any space with a metric on it **can have a topology defined in terms of the metric**. In this section **we define a metric and use it to create the “metric topology” on a set**.

Definition

A **metric** on a set X is a function $d : X \times X \rightarrow \mathbb{R}$ having the following properties:

- (1)** $d(x, y) \geq 0$ for all $x, y \in X$ and $d(x, y) = 0$ if and only if $x = y$.
- (2)** $d(x, y) = d(y, x)$ for all $x, y \in X$.
- (3) (The Triangle Inequality)** $d(x, z) \leq d(x, y) + d(y, z)$ for all $x, y, z \in X$.

- The real number $d(x, y) \geq 0$ is the **distance between x and y** .
- A set X together with a metric d form a **metric space**.
- Define the set $B_d(x, \varepsilon) = \{y \mid d(x, y) < \varepsilon\}$, called the **ε -ball centered at x** – often simply denoted $B(x, \varepsilon)$.

Lemma

Let (X, d) be a metric space and $B_d(x, \varepsilon)$ the ε -ball centered at $x \in X$. Let $y \in B_d(x, \varepsilon)$. Then **there is $\delta > 0$ such that $B_d(y, \delta) \subset B_d(x, \varepsilon)$** .

Proof

Define $\delta = \varepsilon - d(x, y)$ (> 0 since $y \in B_d(x, \varepsilon)$). Then for $z \in B_d(y, \delta)$ we have, by the Triangle Inequality:

$$d(x, z) \leq d(x, y) + d(y, z) < d(x, y) + \delta = d(x, y) + \varepsilon - d(x, y) = \varepsilon.$$

So $y \in B_d(x, \varepsilon)$ and $B_d(y, \delta) \subset B_d(x, \varepsilon)$. ■

Lemma

Let (X, d) be a metric space. Then the collection of all ε -balls $B_d(x, \varepsilon)$ for $x \in X$ and $\varepsilon > 0$ is a basis for a topology on X . This topology is called the *metric topology induced by d* .

Proof.

First, every element $x \in X$ is in a basis element, say $B(x, 1)$. Second, let B_1 and B_2 be two basis elements and let $y \in B_1 \cap B_2$. Then from the previous lemma, there are $\delta_1 > 0$ and $\delta_2 > 0$ with $B(y, \delta_1) \subset B_1$ and $B(y, \delta_2) \subset B_2$. With $\delta = \min \{\delta_1, \delta_2\}$ we then have $B(y, \delta) \subset B_1 \cap B_2$. Since $B(y, \delta)$ is a basis element then the second part of the definition of basis is satisfied. □

The following result is in line with the definition of **open set in a metric space**.

Lemma

A set U is open in the metric topology induced by metric d if and only if for each $y \in U$ there is a $\delta > 0$ such that $B_d(y, \delta) \subset U$.

Proof

Let $y \in U$. By the definition of “topology generated by a basis”, U is open if and only if there is a basis element $B_d(x, \varepsilon)$ with $y \in B_d(x, \varepsilon) \subset U$. Now, by the previous lemma on slide 22, there is $B_d(y, \delta)$ a basis element with $y \in B_d(y, \delta) \subset B_d(x, \varepsilon) \subset U$. ■

Example

For set X , define the metric

$$d(x, y) = \begin{cases} 1 & \text{if } x \neq y \\ 0 & \text{if } x = y \end{cases}$$

(this is in fact a metric). The associated topology is the **discrete topology** on X .

Example (Euclidean spaces $\mathbb{R}^n$)

Definition. Given $\mathbf{x} = (x_1, x_2, \dots, x_n) \in \mathbb{R}^n$, define the **Euclidean norm** of $\mathbf{x}$ as

$$\|\mathbf{x}\| = (x_1^2 + x_2^2 + \dots + x_n^2)^{1/2}.$$

Define the **Euclidean metric** on $\mathbb{R}^n$ as

$$d(\mathbf{x}, \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\| = \left((x_1 - y_1)^2 + (x_2 - y_2)^2 + \dots + (x_n - y_n)^2 \right)^{1/2}.$$

Define the **square metric** ρ as

$$\rho(\mathbf{x}, \mathbf{y}) = \max \{ |x_1 - y_1|, |x_2 - y_2|, \dots, |x_n - y_n| \}$$

We prove that the Euclidean metric actually is a metric in the tutorials. The square metric clearly satisfies the first two parts of the definition of metric. For the Triangle Inequality, notice that for any $x_i, y_i, z_i \in \mathbb{R}$ we have by the Triangle Inequality on $\mathbb{R}$, $|x_i - z_i| \leq |x_i - y_i| + |y_i - z_i|$. So for any $\mathbf{x}, \mathbf{y}, \mathbf{z} \in \mathbb{R}^n$ we have $|x_i - z_i| \leq \rho(\mathbf{x}, \mathbf{y}) + \rho(\mathbf{y}, \mathbf{z})$ for $i = 1, 2, \dots, n$. Hence

$$\rho(\mathbf{x}, \mathbf{z}) = \max_{1 \leq i \leq n} \{|x_i - z_i|\} \leq \rho(\mathbf{x}, \mathbf{y}) + \rho(\mathbf{y}, \mathbf{z}),$$

and the Triangle Inequality holds for ρ . Therefore, ρ is a metric on $\mathbb{R}^n$.

Example (The complex plane $\mathbb{C}$)

The complex plane $\mathbb{C}$ can be identified as a set with the Euclidean plane $\mathbb{R}^2$. If z is a complex number then there exist unique real numbers a and b such that $z = a + ib$. a and b are called the real part and the imaginary part of z and are denoted by $\operatorname{Re}(z)$ and $\operatorname{Im}(z)$. Two complex numbers are said to be equal if their real and imaginary parts are equal:

$$a + ib = c + id \Leftrightarrow a = c \text{ and } b = d$$

We add (or subtract) two complex numbers by adding (or subtracting) their real and imaginary parts, and we multiply them by multiplying them out as in elementary algebra and replacing i^2 by -1 wherever it appears:

$$(a + ib) \pm (c + id) = (a \pm c) + i(b \pm d)$$

and

$$(a + ib)(c + id) = ac + iad + ibc + i^2bd = (ac - bd) + i(ad + bc).$$

Example (The complex plane $\mathbb{C}$ (Continued))

If $z = a + ib$ is a complex number, then its negative $-z$ and its conjugate $\bar{z}$ are defined by $-z = -a - ib$ and $\bar{z} = a - ib$. Simple calculations show directly that

$$\overline{z_1 + z_2} = \bar{z}_1 + \bar{z}_2, \quad \overline{z_1 z_2} = \bar{z}_1 \cdot \bar{z}_2, \quad \text{and} \quad \bar{\bar{z}} = z.$$

The origin, or zero, is the complex number $0 = 0 + i0$. The ordinary distance from $z = a + ib$ to the origin is defined by

$$|z| = (a^2 + b^2)^{1/2}.$$

$|z|$ is called the absolute value or modulus of z , and it is easy to see that $|\bar{z}| = |z|$ and $|z|^2 = z\bar{z}$. The usual metric on $\mathbb{C}$ is defined by

$$d(z_1, z_2) = |z_1 - z_2|.$$

Example (The complex plane $\mathbb{C}$ (Continued))

d is a metric as a consequence of the following properties of the real function $|z|$:

- (i) $|z| \geq 0$ and $|z| = 0 \Leftrightarrow z = 0$.
- (ii) $|-z| = |z|$.
- (iii) $|z_1 + z_2| \leq |z_1| + |z_2|$.

These properties will motivate later on the introduction of a more general class of spaces that we shall call **normed** and **Banach** spaces.

Exercise (Metrics in subspaces)

Let (X, d) be a metric space and let $Y \subset X$ be a subset. Prove that

- (i) The map $d_Y = d|_{Y \times Y} : Y \times Y \longrightarrow \mathbb{R}$ obtained by restricting the metric d to Y is a metric on Y .
- (ii) The metric topology generated by d_Y on Y coincides with the subspace topology induced by the metric topology of (X, d) on Y .

Hint: Prove first that $\forall y \in Y$ and $\forall r > 0$, $B_{d_Y}(y, r) = B_d(y, r) \cap Y$.

Definition

Let X be a topological space. X is said to be **metrizable** if there exists a **metric d on a set X that induces the topology of X** . A metric space is a metrizable space X with a specific metric d that gives the topology of X .

Metrizability is a problem of **fundamental importance in topology**. One of our goals will be to find conditions for that; they are expressed there in the famous theorem called **Urysohn's metrization theorem**. Although the metrizability problem is an important problem in topology, the study of metric spaces as such belongs rather to analysis.

Warning: Metrizability of a space depends only on the topology but properties that involve a specific metric for X in general do not. For instance, one can make the following definition in a metric space:

Definition

Let X be a metric space with metric d . A subset A of X is **bounded** if there is some number M such that $d(a_1, a_2) \leq M$ for every pair $a_1, a_2 \in A$. If A is bounded and nonempty, the **diameter** of A is

$$\text{diam}(A) = \sup \{d(a_1, a_2) \mid a_1, a_2 \in A\}.$$

The next result gives a metric for which every set is bounded.

Theorem

Let X be a metric space with metric d . Define $\bar{d} : X \times X \rightarrow \mathbb{R}$ by $\bar{d}(x, y) = \min\{d(x, y), 1\}$. Then $\bar{d}$ is a metric that induces the same topology as d .

Proof of the theorem

It is easy to see the first two parts of the definition of metric are satisfied by $\bar{d}$. For the third part, we need to confirm the Triangle Inequality:

$$\bar{d}(x, z) \leq \bar{d}(x, y) + \bar{d}(y, z).$$

We consider two cases.

Case 1 . If either $d(x, y) \geq 1$ or $d(y, z) \geq 1$ then the right side of this inequality is at least 1 and so the inequality holds.

Case 2 . If both $d(x, y) < 1$ and $d(y, z) < 1$ then

$$\begin{aligned} d(x, z) &\leq d(x, y) + d(y, z) \text{ by the Triangle Inequality for } d \\ &= \bar{d}(x, y) + \bar{d}(y, z). \end{aligned}$$

Since $\bar{d}(x, z) < d(x, z)$, we have the Triangle Inequality for $\bar{d}$. So $\bar{d}$ is in fact a metric.

Proof of the theorem (continued)

Let $\mathcal{B}$ be the basis for the topology induced by d and let $\mathcal{B}'$ be the basis for the topology induced by $\bar{d}$. By a preceding Lemma (Chap. 2, slide 15), the topology generated by a basis consists of all unions of basis elements. Notice that $\mathcal{B}'$ consists of all ε -balls where $\varepsilon \leq 1$, so $\mathcal{B}' \subset \mathcal{B}$ and the topology generated by $\mathcal{B}'$ is a subset of the topology generated by $\mathcal{B}$.

But for any $B_d(x, \varepsilon) \in \mathcal{B}$ we know that $B_d(x, \varepsilon)$ can be written as a union over all elements of $B_d(x, \varepsilon)$ of balls centered at these elements with radius less than 1 (here we use the Lemma on slide 24):

$$B_d(x, \varepsilon) = \bigcup_{y \in B_d(x, \varepsilon)} B_d(y, \delta_y)$$

where $\delta_y = \min\{\varepsilon, 1\}$ and $B_d(y, \delta_y) \subset B_d(x, \varepsilon)$ as in Lemma on slide 24. So every set in the topology generated by $\mathcal{B}$ is also in the topology generated by $\mathcal{B}'$ (since $B_d(y, \delta_y) \in \mathcal{B}'$ since $\delta_y \leq 1$). So the topologies are the same and d and $\bar{d}$ induce the same topology on X . ■

The topologies on $\mathbb{R}^n$ induced by the Euclidean metric and the square metric are the same as the product topology on $\mathbb{R}^n$. To establish this we need a preliminary result.

Lemma

Let d and d' be two metrics on the set X . Let $\mathcal{T}$ and $\mathcal{T}'$ be the topologies they induce, respectively. Then $\mathcal{T}'$ is finer than $\mathcal{T}$ ($\mathcal{T}' \supset \mathcal{T}$) if and only if for each $x \in X$ and each $\varepsilon > 0$, there exists a $\delta > 0$ such that $B_{d'}(x, \delta) \subset B_d(x, \varepsilon)$.

Proof of the lemma

Suppose that $\mathcal{T}'$ is finer than $\mathcal{T}$. Let $B_d(x, \varepsilon)$ be a basis element for the metric topology $\mathcal{T}$. By a preceding Lemma (Chap. 2, slide 19) (the $(1) \Rightarrow (2)$ part) there is a basis element such that $x \in B' \subset B_d(x, \varepsilon)$. By the lemma in page 24, there is $B_{d'}(x, \delta) \subset B' \subset B_d(x, \varepsilon)$ and the first claim holds.

Suppose the ε - δ condition holds. Given a basis element B for the metric topology for $\mathcal{T}$ containing x , by the lemma in page 24 there is a basis element $B_d(x, \varepsilon) \subset B$. By the hypothesized ε - δ condition there is $B' = B_{d'}(x, \delta) \subset B_d(x, \varepsilon) \subset B$. By the Lemma (Chap. 2, slide 19) (the $(2) \Rightarrow (1)$ part), $\mathcal{T}'$ is finer than $\mathcal{T}$. ■

Theorem

The topologies on $\mathbb{R}^n$ induced by the Euclidean metric d and the square metric ρ are the same as the product topology on $\mathbb{R}^n$.

Closed balls

Definition

Let d be a metric on the set X and let $\epsilon \geq 0$. The **closed ball** $B_d[x_0, \epsilon]$ with center x_0 and radius r is the subset of X defined by

$$B_d[x_0, \epsilon] = \{x \in X \mid d(x, x_0) \leq r\}$$

$B_d[x_0, \epsilon]$ contains its center and when $r = 0$ it contains only its center. The closed balls on the real line are precisely the closed intervals. The following theorem justifies the adjective in the phrase “**closed** ball.”

Theorem

*In any metric space X , **each closed ball is a closed set.***

Proof

Let $B_d[x_0, \epsilon]$ be a closed ball in X . It suffices to show that its complement $X \setminus B_d[x_0, \epsilon]$ is open. $X \setminus B_d[x_0, \epsilon]$ is open if it is empty, so we may assume that it is non-empty.

Let x be a point in $X \setminus B_d[x_0, \epsilon]$. Since $d(x, x_0) > \epsilon$, $\epsilon_1 = d(x, x_0) - \epsilon$ is a positive real number. We take ϵ_1 as the radius of an open ball $B_d(x, \epsilon_1)$ centered on x , and we show that $X \setminus B_d[x_0, \epsilon]$ is open by showing that $B_d(x, \epsilon_1) \subseteq X \setminus B_d[x_0, \epsilon]$. Let y be a point in $B_d(x, \epsilon_1)$, so that $d(y, x) < \epsilon_1$. On the basis of this and the fact that $d(x_0, x) \leq d(x_0, y) + d(y, x)$, we see that

$$\begin{aligned} d(y, x_0) &\geq d(x, x_0) - d(y, x) > d(x, x_0) - \epsilon_1 \\ &= d(x, x_0) - [d(x, x_0) - \epsilon] = \epsilon, \end{aligned}$$

so that y is in $X \setminus B_d[x_0, \epsilon]$. ■

Important observation

One is inclined to believe that the closure $\overline{B_d(x_0, \epsilon)}$ of the open ball $B_d(x_0, \epsilon)$ is the closed ball $B_d[x_0, \epsilon]$. This **is not necessarily true**. Take the metric defined in the previous example

$$d(x, y) = \begin{cases} 1 & \text{if } x \neq y \\ 0 & \text{if } x = y \end{cases}$$

In this case and for any $x_0 \in X$, $B_d(x_0, 1) = \{x_0\} = \overline{B_d(x_0, 1)}$, while $B_d[x_0, 1] = X$. This statement **holds nevertheless true** for the structure that we introduce in the next section, that is, **normed spaces**.

Metrics induced by norms

The examples provided above with $\mathbb{R}^n$ and $\mathbb{C}$ are particular cases of **normed linear spaces**.

Definition

A **normed linear space** $(V, \|\cdot\|)$ is a vector space V on which there is defined a **norm**, that is, a function $\|\cdot\| : V \longrightarrow [0, \infty)$ which assigns to each element x in the space a real number $\|x\|$ in such a manner that

- (1)** $\|x\| \geq 0$, and $\|x\| = 0 \Leftrightarrow x = 0$;
- (2)** $\|x + y\| \leq \|x\| + \|y\|$;
- (3)** $\|\alpha x\| = |\alpha| \|x\|$.

A normed linear space is simply a linear space in which there is available a notion of the distance from an arbitrary element to the origin.

Lemma

A *normed linear space* is a metric space with respect to the induced metric defined by

$$d(x, y) = \|x - y\|.$$

An important particular case of normed spaces are the **Banach spaces** that we introduce later on. A Banach space is a normed linear space with an additional **completeness** condition that we define later.

Example

The **Euclidean metric** in $\mathbb{R}^n$ is induced by the **Euclidean norm**. The **square metric** in $\mathbb{R}^n$ is induced by the **infinity** or the **maximum norm** $\|\cdot\|_\infty$ defined by

$$\|\mathbf{x}\|_\infty = \max_{1 \leq i \leq n} \{|x_i|\}.$$

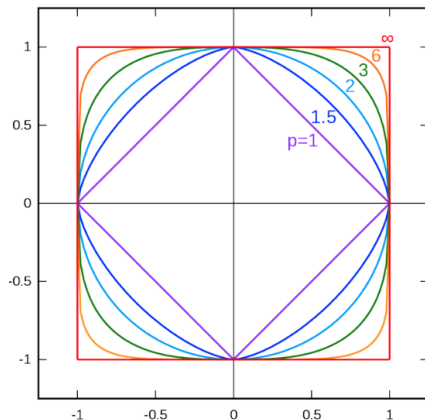
The **usual metric** on $\mathbb{C}$ is induced by the **modulus function** $|\cdot| : \mathbb{C} \rightarrow [0, \infty)$, which is a **norm** in itself.

Example (p -norms in $\mathbb{R}^n$)

The previous example is a particular case of a more general family of norms that can be introduced in $\mathbb{R}^n$. Indeed, for any $p \geq 1$, the p -norm (also called ℓ_p -norm) of vector $\mathbf{x} = (x_1, \dots, x_n)$ is defined by

$$\|\mathbf{x}\|_p = \left(\sum_{i=1}^n |x_i|^p \right)^{1/p}$$

For $p = 1$, we get the so-called **taxicab norm**, for $p = 2$, we get the **Euclidean norm**, and as p approaches ∞ the balls of p -norm approach those of the **infinity** or **maximum** norm.



These pictures show that all these norms induce the same topologies in $\mathbb{R}^2$.

A first infinite-dimensional example

Example (Sequence spaces)

Denote by $\mathbf{x} = (x_1, x_2, \dots) : \mathbb{N} \longrightarrow \mathbb{R}$ a sequence with real entries. The space $\mathbb{R}^{\mathbb{N}}$ of such sequences forms an infinite dimensional vector space with the operations:

$$\begin{aligned}\mathbf{x} + \mathbf{y} &= (x_1 + y_1, x_2 + y_2, \dots) \\ \lambda \cdot \mathbf{x} &= (\lambda x_1, \lambda x_2, \dots).\end{aligned}$$

For any $1 \leq p < \infty$, define the norm $\|\cdot\|_p : \mathbb{R}^{\mathbb{N}} \longrightarrow [0, \infty)$ by

$$\|\mathbf{x}\|_p = \left(\sum_{n \in \mathbb{N}} |x_n|^p \right)^{1/p}.$$

Example (Sequence spaces (continued))

Define by ℓ^p as the vector subspace of $\mathbb{R}^{\mathbb{N}}$ determined by the condition $\|\mathbf{x}\|_p < \infty$, that is:

$$\ell^p = \{\mathbf{x} \in \mathbb{R}^{\mathbb{N}} \mid \|\mathbf{x}\|_p < \infty\}$$

Analogously, define

$$\|\mathbf{x}\|_{\infty} = \sup_{n \in \mathbb{N}} \{|x_n|\} \quad \text{and} \quad \ell^{\infty} = \{\mathbf{x} \in \mathbb{R}^{\mathbb{N}} \mid \|\mathbf{x}\|_{\infty} < \infty\}$$

For any $1 \leq p \leq \infty$, the vector subspaces ℓ^p are normed spaces. ℓ^{∞} contains three important closed vector subspaces, namely:

- $\mathbf{c}$ is the subspace of **convergent sequences**.
- $\mathbf{c}_0$ is the subspace of **convergent sequences with zero limit**.

Outline

- 1 Continuous functions
- 2 The metric topology
- 3 Continuity in metric spaces**
- 4 More about topology on product spaces
- 5 References

Continuity in metric spaces

In this section we give a number of results for metric spaces which are familiar from calculus and real analysis.

The following theorem shows that the usual ε - δ definition of continuity is equivalent to our definition of continuity in terms of inverse images of open sets when we have a metric.

Theorem

Let $f : X \rightarrow Y$. let X and Y be metrizable with metrics d_X and d_Y , respectively. Then continuity of f is equivalent to the requirement that given $x \in X$ and given $\varepsilon > 0$, there exists $\delta > 0$ such that

$$d_X(x, y) < \delta \Rightarrow d_Y(f(x), f(y)) < \varepsilon$$

Proof

Suppose f is continuous. Let $x \in X$ and $\varepsilon > 0$ be given. Consider the set $f^{-1}(B(f(x), \varepsilon))$. Since f is continuous then by definition of continuity, $f^{-1}(B(f(x), \varepsilon))$ is open in X since $B(f(x), \varepsilon)$ is open in Y . Since $x \in f^{-1}(B(f(x), \varepsilon))$ then by the lemma in page 22 there is $\delta > 0$ such that $B(x, \delta) \subset f^{-1}(B(f(x), \varepsilon))$. Then $d_X(x, y) < \delta$ implies $y \in B(x, \delta)$, so $f(y) \in B(f(x), \varepsilon)$ and $d_Y(f(x), f(y)) < \varepsilon$.

Conversely, suppose that the ε - δ condition is satisfied. Let V be an open set in Y . Let us show that $f^{-1}(V)$ is open. Given $x \in f^{-1}(V)$, we have $f(x) \in V$. Since V is open and $f(x) \in V$ then by the lemma in page 24 there is $\varepsilon > 0$ such that $B(f(x), \varepsilon) \subset V$. By the ε/δ hypothesis, there is $\delta > 0$ such that $d_X(x, y) < \delta$ implies $d_Y(f(x), f(y)) < \varepsilon$ (i.e., $f(y) \in B(f(x), \varepsilon)$). In other words: $f(B(x, \delta)) \subset B(f(x), \varepsilon)$. So $x \in B(x, \delta) \subset f^{-1}(B(f(x), \varepsilon)) \subset V$. Therefore, by the lemma on slide 24, $f^{-1}(V)$ is open and so f is continuous. ■

Bounded maps

When the range of a map is a metric space we can talk about **boundedness of that map**.

Definition

Let X be a set and let (Y, d) be a metric space. We say that a function $f : X \rightarrow Y$ is **bounded** if its image $f(X)$ is a **bounded subset of the metric space (Y, d)** . The **set of bounded functions** between X and (Y, d) is denoted by $\mathcal{B}(X, Y)$.

Example

The function $\sin : \mathbb{R} \rightarrow \mathbb{R}$ is bounded while $\exp : \mathbb{R} \rightarrow \mathbb{R}$ is not when $\mathbb{R}$ is endowed with the standard metric.

Example (Another infinite dimensional normed space)

Let X be a set and let $\mathbb{R}$ considered as a normed space with the absolute value. The set $\mathcal{B}(X, \mathbb{R})$ forms an infinite-dimensional vector space with the operations defined by:

$$\begin{aligned}(f + g)(x) &= f(x) + g(x), \\ (\lambda f)(x) &= \lambda \cdot f(x).\end{aligned}$$

The map $\|\cdot\|_{\infty} : \mathcal{B}(X, \mathbb{R}) \longrightarrow [0, \infty)$ defined by

$$\|f\|_{\infty} = \sup_{x \in X} \{|f(x)|\}$$

makes it into a normed space. (See [Simmons, Exercise 9.5])

Convergence in metric spaces: The Sequence Lemma

Recall from real and complex analysis that a **closed set contains its limit points**. The following result for topological spaces is related to this familiar property from $\mathbb{R}$.

Lemma (The Sequence Lemma)

Let X be a topological space. Let $A \subset X$. If there is a sequence of points of A converging to x , then $x \in \bar{A}$. If X is metrizable and $x \in \bar{A}$ then there is a sequence $\{x_n\} \subset A$ such that $\{x_n\} \rightarrow x$.

Proof the Sequence Lemma

Suppose that $\{x_n\} \rightarrow x$ where $\{x_n\} \subset A$. Then any given neighborhood U of x , there is, by the definition of limit of a sequence, $N \in \mathbb{N}$ such that $x_n \in U$ for all $n \geq N$. So every neighborhood of x contains an element of the sequence and hence an element of set A . Then by Theorem (Chap. 2, slide 56) (part (a)), $x \in \bar{A}$.

Conversely, suppose that X is metrizable and $x \in \bar{A}$. Let d be a metric for the topology of X . For each $n \in \mathbb{N}$, consider $B_d(x, 1/n)$. This is an open set containing x and so by the Theorem (Chap. 2, slide 56) (part (a)), $B(x, 1/n)$ contains an element of A , say x_n . Then consider the resulting sequence $\{x_n\}$. We shall show that it converges to x .

Given an open set U which contains x , there is $\varepsilon > 0$ such that $B_d(x, \varepsilon) \subset U$, by the lemma on slide 25. Since $\varepsilon > 0$, then for some $N \in \mathbb{N}$ we have $1/N < \varepsilon$. So by construction of the sequence, for all $n \geq N$ we have $x_n \in B(x, 1/n) \subset B(x, 1/N) \subset B(x, \varepsilon) \subset U$. So $\{x_n\} \rightarrow x$ by the definition of convergent sequence. ■

Limit points and sequence limits in metric spaces

The following result (see [Simmons, page 72]) relates the two concepts introduced in the previous chapter, in the context of metric spaces, namely those of **limit point of a set** and **limit of a sequence**.

Theorem

*If a convergent sequence in a metric space **has infinitely many distinct points**, then **its limit is a limit point of the set of points of the sequence**.*

Proof of the theorem

Let $\{x_n\}$ be a convergent sequence in the metric space X with limit x . We assume that x is not a limit point of the set of points of the sequence, and we show that it follows from this that the sequence has only finitely many distinct points. The assumption implies that there exists an open ball $B(x, r)$ centered on x which contains no point of the sequence different from x . However, since x is the limit of the sequence, all x_n 's from some place on must lie in $B(x, r)$, hence must coincide with x . From this there are only finitely many distinct points in the sequence. ■

Continuity with sequences

Recall from the real and complex analysis classes that a function f is continuous at point x if and only if every sequence $\{x_n\} \rightarrow x$ in the domain of f satisfies $\{f(x_n)\} \rightarrow f(x)$.

Theorem

Let $f : X \rightarrow Y$. If f is continuous then for every convergent sequence $\{x_n\} \rightarrow x$ in X , the sequence $\{f(x_n)\} \rightarrow f(x)$ in Y . If X is metrizable and for any sequence $\{x_n\} \rightarrow x$ in X we have $\{f(x_n)\} \rightarrow f(x)$ in Y , then f is continuous.

Proof of the theorem

Suppose f is continuous and let $\{x_n\} \rightarrow x$ in X . Let V be a neighborhood of $f(x)$. Then $f^{-1}(V)$ is open and contains x . Since $\{x_n\} \rightarrow x$, by the definition of convergent sequence, there is $N \in \mathbb{N}$ such that for all $n \geq N$ we have $x_n \in f^{-1}(V)$. Then $f(x_n) \in V$ for all $n \geq N$ and so (by definition again) $\{f(x_n)\} \rightarrow f(x)$.

Conversely, suppose X is metrizable and suppose for any $x \in X$ and any sequence $\{x_n\} \rightarrow x$ in X we have $\{f(x_n)\} \rightarrow f(x)$.

We shall use the Theorem slide 9 (the (2) $\Rightarrow$ (1) part: if for every subset A of X , one has $f(\bar{A}) \subset \overline{f(A)}$, then f is continuous).

Let $A \subset X$. If $x \in \bar{A}$ then there is a sequence $\{x_n\} \subset A$ such that $\{x_n\} \rightarrow x$ by the Sequence Lemma (part 2 ; needs metrizability of X). By hypothesis, $\{f(x_n)\} \rightarrow f(x)$. Since $\{x_n\} \subset A$ then $f(x) \in \overline{f(A)}$ by the Sequence Lemma (part 1; notice that this does not require the metrizability of Y). Since $x \in \bar{A}$ is arbitrary, then $f(\bar{A}) \subset \overline{f(A)}$. Hence, f is continuous. ■

First countability

The hypotheses of metrizability in the proofs of the second parts of the Sequence Lemma and the Theorem on slide 57 can be weakened. In this direction, we have the following definitions.

Definition

A topological space X is said to **have a countable basis** at the point x if there is a countable collection $\{U_n\}_{n \in \mathbb{N}}$ of neighborhoods of x such that any neighborhood U of x contains at least one of the sets U_n . A space that has a countable basis at each of its points is said to satisfy the **first countability axiom**.

In the proof of the Sequence Lemma we can replace the open balls $B_d(x, 1/n)$ with the intersection of the first n elements of a countable basis at x , $B_n = U_1 \cap U_2 \cap \cdots \cap U_n$. So we can change the hypothesis from “metrizable” to “first countable” and the Sequence Lemma and the Theorem on slide 57 still hold (with no modifications to the proof of the Theorem).

Every metrizable space is first countable, simply take as a countable basis at x the open balls $B_d(x, 1/n)$ for $n \in \mathbb{N}$. The countability axioms are considered in more detail in Section 30 of [Munkres].

Lemma

The *addition, subtraction, and multiplication* operations are *continuous* from $\mathbb{R} \times \mathbb{R}$ into $\mathbb{R}$, and the quotient operation is a continuous function from $\mathbb{R} \times (\mathbb{R} \setminus \{0\})$ into $\mathbb{R}$.

Theorem

If X is a topological space and if $f, g : X \rightarrow \mathbb{R}$ are continuous, then $f + g$, $f - g$, and $f \cdot g$ are *continuous*. If $g(x) \neq 0$ for all $x \in X$ then f/g is continuous.

Corollary

Let X be a topological space and let $\mathcal{C}_B(X, \mathbb{R}) = \mathcal{C}(X, \mathbb{R}) \cap \mathcal{B}(X, \mathbb{R})$ be the space of *continuous and bounded real valued functions*. Then $\mathcal{C}_B(X, \mathbb{R})$ is a vector space and the norm $\|\cdot\|_\infty : \mathcal{C}_B(X, \mathbb{R}) \longrightarrow [0, \infty)$

$$\|f\|_\infty = \sup_{x \in X} \{|f(x)|\}$$

makes it into a normed space.

Uniform convergence

We now define the **uniform convergence** of a sequence of functions. Recall that the uniform limit of a sequence of Riemann integrable functions $\{f_n\}$ on $[a, b]$ is Riemann integrable on $[a, b]$ and

$$\lim_{n \rightarrow \infty} \left(\int_a^b f_n(x) dx \right) = \int_a^b \left(\lim_{n \rightarrow \infty} f_n(x) \right) dx.$$

Definition

Let $f_n : X \rightarrow Y$ be a sequence of functions from set X to metric space Y . Let d be the metric for Y . The sequence of functions $\{f_n\}$ **converges uniformly** to the function $f : X \rightarrow Y$ if given $\varepsilon > 0$ there is $N \in \mathbb{N}$ such that $d(f_n(x), f(x)) < \varepsilon$ **for all $n > N$ and for all $x \in X$.**

The “**uniform**” in **uniform convergence** is the fact that for a given $\varepsilon > 0$, the same $N \in \mathbb{N}$ “works” **FOR ALL $x \in X$** (that is, the $N \in \mathbb{N}$ works uniformly across set X).

Exercise

Let X be a set and let $f_n : X \longrightarrow \mathbb{R}$ be a sequence of bounded functions in $\mathcal{B}(X, \mathbb{R})$. Prove that the sequence of functions $\{f_n\}$ **converges uniformly** to the function $f : X \rightarrow \mathbb{R}$ **if and only if** $f \in \mathcal{B}(X, \mathbb{R})$ and $f_n \longrightarrow f$ **as a sequence in the metric (normed) space $(\mathcal{B}(X, \mathbb{R}), \|\cdot\|_\infty)$.**

Uniform Limit Theorem

Theorem (Uniform Limit Theorem)

Let $f_n : X \rightarrow Y$ be a sequence of continuous functions from the topological space X to the metric space Y . If $\{f_n\}$ *converges uniformly* to f , then *f is continuous*.

Proof of the Uniform Limit Theorem

Let V be open in Y and let $x_0 \in f^{-1}(V)$. Let $y_0 = f(x_0) \in V$ and choose $\varepsilon > 0$ such that $B(y_0, \varepsilon) \subset V$ (by Lemma page 24).

We want to find U neighborhood of x_0 such that $f(U) \subset B(f(x_0), \varepsilon) \subset V$.

Since $\{f_n\}$ converges uniformly to f on X then there is $N \in \mathbb{N}$ such that for all $n \geq N$ and for all $x \in X$ we have

$$d(f_n(x), f(x)) < \varepsilon/3. \quad (*)$$

Since f_N is continuous, there is a neighborhood U of x_0 such that

$$f_N(U) \subset B(f_N(x_0), \varepsilon/3) \quad (**)$$

by Theorem on slide 9 (the $(1) \Rightarrow (4)$ part where $B(f_N(x_0), \varepsilon/3)$ is treated as a neighborhood of $f(x_0)$).

Proof of the Uniform Limit Theorem (continued)

Next, if $x \in U$ then

$$d(f(x), f_N(x)) < \varepsilon/3 \text{ by } (*) \text{ with } n = N$$

$$d(f_N(x), f_N(x_0)) < \varepsilon/3 \text{ by } (**) \text{ since } x \in U$$

$$d(f_N(x_0), f(x_0)) < \varepsilon/3 \text{ by } (*) \text{ with } n = N \text{ and } x = x_0.$$

Then by the Triangle Inequality,

$$\begin{aligned} d(f(x), f(x_0)) &\leq d(f(x), f_N(x)) + d(f_N(x), f_N(x_0)) + d(f_N(x_0), f(x_0)) \\ &< \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} = \varepsilon \end{aligned}$$

for all $x \in U$. So U is a neighborhood of x_0 with

$f(U) \subset B(f(x_0), \varepsilon) \subset V$. So by Theorem on slide 9 (the (4) $\Rightarrow$ (1) part), f is continuous. ■

Exercise

Show that the previous theorem implies that the vector subspace $(\mathcal{C}_B(X, \mathbb{R}), \|\cdot\|_\infty)$ is **closed** in $(\mathcal{B}(X, \mathbb{R}), \|\cdot\|_\infty)$.

Hint: Use the Sequence Lemma.

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- 2 The metric topology
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More about topology on product spaces

In the last chapter we defined the product topology on the product of two topological spaces X and Y . In this section we consider **arbitrary products of topological spaces** and give **two topologies** on these spaces, the **box topology** and the **product topology**. For finite products, these topologies coincide.

General (Cartesian) product spaces

Definition

Let $\{A_\alpha\}_{\alpha \in J}$ be an indexed family of sets, where J is an (arbitrary) **index set**. The **Cartesian product** of this indexed family, denoted by $\prod_{\alpha \in J} A_\alpha$, is defined to be the set of all J -tuples $(x_\alpha)_{\alpha \in J}$ of elements of X such that $x_\alpha \in A_\alpha$ for each $\alpha \in J$. That is, $\prod_{\alpha \in J} A_\alpha$ is the **set of all functions $\mathbf{x} : J \rightarrow \cup_{\alpha \in J} A_\alpha$ such that $\mathbf{x}(\alpha) \in A_\alpha$ for each $\alpha \in J$** .

$$J = \{1, \dots, n\} \quad \leadsto \quad \prod_{\alpha \in J} A_\alpha = A_1 \times \dots \times A_n.$$

The box topology

Definition

Let $\{X_\alpha\}_{\alpha \in J}$ be an indexed family of topological spaces. We take as a basis for a topology on the product space $\prod_{\alpha \in J} X_\alpha$ the collection of all sets of the form $\prod_{\alpha \in J} U_\alpha$ where U_α is open in X_α for each $\alpha \in J$. The topology generated by this basis is called the **box topology**.

When $J = \{1, 2\}$, the box topology is the same as the product topology on $X \times Y$ from Chapter 1.

The collection of all sets of the form $\prod_{\alpha \in J} U_{\alpha}$ do, in fact, form a basis for a topology on $\prod_{\alpha \in J} X_{\alpha}$. First, $\prod_{\alpha \in J} X_{\alpha}$ itself is such a set so each $\mathbf{x} \in \prod_{\alpha \in J} X_{\alpha}$ is in some basis element. Second, suppose $\mathbf{x} \in (\prod_{\alpha \in J} U_{\alpha}) \cap (\prod_{\alpha \in J} V_{\alpha}) = \prod_{\alpha \in J} U_{\alpha} \cap V_{\alpha}$. Notice that each U_{α}, V_{α} is open in X_{α} , so $\prod_{\alpha \in J} U_{\alpha} \cap V_{\alpha}$ is a basis element and the definition of basis is satisfied.

The product topology

Definition

Define the **projection** $\pi_\beta : \prod_{\alpha \in J} X_\alpha \rightarrow X_\beta$ for $\beta \in J$ as the function assigning to each element of $\prod_{\alpha \in J} X_\alpha$ its β th coordinate, $\pi_\beta((x_\alpha)_{\alpha \in J}) = x_\beta$. Let $\mathcal{S}_\beta$ denote the collection

$$\mathcal{S}_\beta = \left\{ \pi_\beta^{-1}(U_\beta) \mid U_\beta \text{ is open in } X_\beta \right\}$$

and let $\mathcal{S}$ denote the union of these collections, $\mathcal{S} = \cup_{\beta \in J} \mathcal{S}_\beta$. The **topology generated by the subbasis $\mathcal{S}$** is **the product topology** and in this topology $\prod_{\alpha \in J} X_\alpha$ is called the **product space**.

In order to verify that $\mathcal{S}$ really is a subbasis for a topology, we must only show that there are sets in $\mathcal{S}$ which union to give $\prod_{\alpha \in J} X_{\alpha}$ (by the definition of subbasis). Since the total set X_{β} is an open set for any topology in X_{β} and $\pi_{\beta}^{-1}(X_{\beta}) = \prod_{\alpha \in J} X_{\alpha} \in \mathcal{S}$ so, in fact, $\mathcal{S}$ really is a subbasis for a topology.

Theorem (Comparison of the Box and Product Topologies)

The **box topology** on $\prod_{\alpha \in J} X_{\alpha}$ has as a basis all sets of the form $\prod_{\alpha \in J} U_{\alpha}$ where U_{α} is open in X_{α} for each $\alpha \in J$.

The **product topology** on $\prod_{\alpha \in J} X_{\alpha}$ has as a basis all sets of the form $\prod_{\alpha \in J} U_{\alpha}$ where U_{α} is open in X_{α} for each $\alpha \in J$ and $U_{\alpha} = X_{\alpha}$ except for finitely many values of α .

If J is a finite set then the box topology and the product topology on $\prod_{\alpha \in J} X_{\alpha}$ coincide since they have bases with the same elements. In general the box topology is a finer topology than the product topology on $\prod_{\alpha \in J} X_{\alpha}$.

Because the product topology is coarser (or “weaker”) than the box topology, **more “desirable” properties hold in the product topology**. This will be illustrated in the next result. For this reason, when considering $\prod_{\alpha \in J} X_{\alpha}$ (which we abbreviate $\prod X_{\alpha}$) **we assume that it has the product topology, unless stated otherwise**.

The identity map:

$$\text{box topology} \rightarrow \text{product topology}$$

is continuous.

Theorem (Products of maps)

*Let $f : A \rightarrow \prod_{\alpha \in J} X_{\alpha}$ be given as $f(a) = (f_{\alpha}(a))_{\alpha \in J}$ where $f_{\alpha} : A \rightarrow X_{\alpha}$ for each $\alpha \in J$. Let $\prod X_{\alpha}$ have the **product topology**. Then the function f is continuous if and only if each functions f_{α} is continuous.*

Proof of the theorem

Let $\pi_\beta : \prod X_\alpha \rightarrow X_\beta$ be the projection mapping. The function π_β is continuous since for open $U_\beta \in X_\beta$ we have $\pi_\beta^{-1}(U_\beta) = \prod Y_\alpha$ where $Y_\alpha = X_\alpha$ for $\alpha \neq \beta$ and $Y_\alpha = U_\alpha$ if $\alpha = \beta$, and this is an element of the subbasis for the product topology and so is open. So for $f : A \rightarrow \prod X_\alpha$ continuous, the function $f_\beta = \pi_\beta \circ f$ is continuous by Theorem (c) on slide 15.

Proof of the theorem (continued)

Conversely, suppose each f_α is continuous for $\alpha \in J$. To prove $f : A \rightarrow \prod X_\alpha$ is continuous, it suffices to prove that the inverse image under f of each subbasis element is open in A (since each open set in $\prod X_\alpha$ is a union of finite intersections of subbasis elements and the inverse image of a union or intersection is the union or intersection of inverse images).

A typical subbasis element for the product topology on $\prod X_\alpha$ is of the form $\pi_\beta^{-1}(U_\beta)$ where $\beta \in J$ and U_β is open in X_β

Now $f^{-1}(\pi_\beta^{-1}(U_\beta)) = (\pi_\beta \circ f)^{-1}(U_\beta) = f_\beta^{-1}(U_\beta)$ since $f_\beta = \pi_\beta \circ f$.

Since f_β is hypothesized to be continuous, then $f_\beta^{-1}(U_\beta)$ is open and so inverse images under f of sets of the form $\pi_\beta^{-1}(U_\beta)$ are open sets. So inverse images under f of subbasis elements (and hence open set under the product topology) are open. That is, f is continuous. ■

Example

Consider $\mathbb{R}^\omega = \prod_{n \in \mathbb{Z}_+} \mathbb{R}$ where each $\mathbb{R}$ has the standard topology and $\mathbb{R}^\omega$ has the box topology. Define $f : \mathbb{R} \rightarrow \mathbb{R}^\omega$ by the equation $f(t) = (t, t, t, \dots)$. Then $f_n(t) = t$ and so $f_n : \mathbb{R} \rightarrow \mathbb{R}$ is continuous for each $n \in \mathbb{N}$ (and so if we put the product topology on $\mathbb{R}^\omega$ then f would be continuous by the previous Theorem).

Consider the basis element for the box topology $B = (-1, 1) \times (-1/2, 1/2) \times (-1/3, 1/3) \times \dots$. Notice that $f^{-1}(B) = (-1, 1) \cap (-1/2, 1/2) \cap (-1/3, 1/3) \cap \dots = \{0\}$. But B is open in $\mathbb{R}^\omega$ and $\{0\}$ is not open in $\mathbb{R}$. So f is not continuous.

Outline

- 1 Continuous functions
- 2 The metric topology
- 3 Continuity in metric spaces
- 4 More about topology on product spaces
- 5 References

References I



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George F. Simmons. Topology and Modern Analysis. McGraw-Hill, 1963.